Phase 11: Statistical Validation Framework & Econometric Formulations

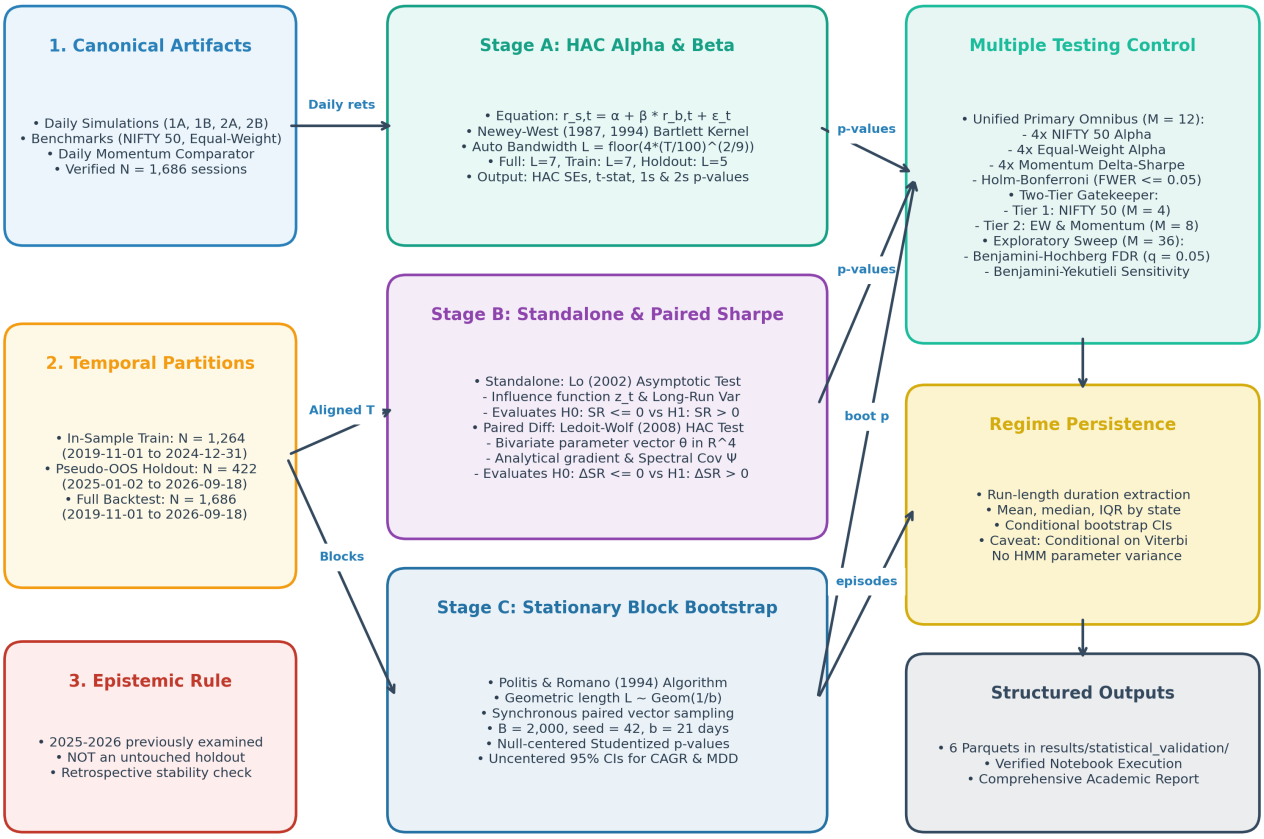
Machine-Learning-Based Sector Rotation Regime Discovery in the Indian Equity Market

Authoritative Research Methodology & Mathematical Specification

Abstract: This document provides the formal mathematical, econometric, and procedural foundation for Phase 11 statistical validation. We establish rigorous hypothesis testing protocols to determine whether the canonical sector rotation strategies (1A, 1B, 2A, 2B) possess genuine outperformance beyond noise. The framework incorporates Newey-West HAC inference under proven ARCH effects, Lo (2002) asymptotic Sharpe tests, Ledoit-Wolf (2008) bivariate comparative tests, Politis-Romano (1994) Stationary Block Bootstrapping, and mathematically closed Multiple-Testing error control via a Unified Omnibus Holm-Bonferroni Family (M = 12). All empirical data boundaries, sample sizes, and parameter conventions are authoritatively codified.

1. Overall Statistical Validation Flowchart

The diagram below outlines the sequential flow of information from frozen Phase 10 simulation outputs through data alignment, econometric estimation, block-bootstrap resampling, and multiple-testing error control to final structured outputs.



2. Dataset Partitions and Epistemic Status

Scientific integrity requires precise characterization of evaluation data. Because the 2025–2026 period was previously inspected in Phases 10.1–10.6, it cannot be described as a fresh, untouched holdout. The three evaluation windows are formally defined as:

Partition Label	Date Range	Sample N	Epistemic Role & Methodological Interpretation
Model In-Sample Development (Train)	2019-11-01 to 2024-12-31	1,264	Model calibration window (HMM fitting, regime threshold tuning). Statistical tests evalu
Previously Inspected Pseudo-OOS (Holdout)	2025-01-02 to 2026-09-18	422	Temporally post-training, but previously observed in Phase 10. Evaluates post-training
Full Historical Lifecycle Backtest (Full)	2019-11-01 to 2026-09-18	1,686	Complete multi-cycle market history (7 years). Primary retrospective track record signi

3. Complete Mathematical Formulations

3.1 Core Backtest Metric Invariants

All strategies and benchmarks adhere strictly to the Phase 10.2 approved backtesting invariants. Net daily returns *r_net,t* incorporate actual transaction costs (15 bps) and portfolio weight drift. Cash earns exactly 0.0% nominal daily yield. The annualization factor is strictly *A = 252.0*.

Compounded Annual Growth Rate (CAGR):

$$CAGR = (NAV_T / NAV_0)^{(252.0 / N)} - 1.0$$

Annualized Realized Volatility:

$$\sigma_{ann} = \sqrt{ (1 / (N - 1)) * \sum_{t=1}^{N} (r_t - mean(r))^2 } * \sqrt{252.0}$$

Annualized Sharpe Ratio (Zero Risk-Free Convention):

$$SR_{ann} = (mean(r) * 252.0) / \sigma_{ann} = (mean(r) / std(r)) * \sqrt{252.0}$$

Maximum Drawdown (MDD):

$$DD_t = (NAV_t - \max_{1 <= s <= t} NAV_s) / (\max_{1 <= s <= t} NAV_s), MDD = \min_{1 <= t <= N} DD_t$$

3.2 Jensen's Alpha & Newey-West (1987, 1994) HAC Regression

Because empirical diagnostics revealed massive volatility clustering across all return series (Engle ARCH LM p < 10^-24), classical OLS standard errors underestimate variance. Single-factor market models are estimated via OLS with Newey-West HAC standard errors:

Single-Factor Regression Equation:

$$r_{strategy,t} = \alpha_{daily} + \beta * r_{benchmark,t} + \varepsilon_t, \quad t = 1, ..., T$$

$$\alpha_{ann} = \alpha_{daily} * 252.0, \quad \beta = Cov(r_{strat}, r_{bench}) / Var(r_{bench})$$

Automatic Bartlett Kernel Bandwidth (Newey & West 1994):

$$L = \text{floor}(4.0 * (T / 100)^{(2/9)}) \text{ --> Full (N=1686): L=7; Train (N=1264): L=7; Holdout (N=422): L=5}$$

$$w(j, L) = 1.0 - (j / (L + 1)), \quad j = 1, ..., L$$

HAC Spectral Covariance Matrix (v_t = x_t * e_t):

$$S_{HAC} = \Gamma_0 + \sum_{j=1}^{L} w(j, L) * (\Gamma_j + \Gamma_j^T), \quad \Gamma_j = (1 / T) * \sum_{t=j+1}^{T} v_t * v_{t-j}^T$$

$$\Omega_{HAC} = (X^T X)^{-1} * (T * S_{HAC}) * (X^T X)^{-1}$$

$$SE_{HAC}(\alpha_{ann}) = 252.0 * \sqrt{ \Omega_{HAC}[0, 0] }, \quad t_\alpha = \alpha_{daily} / \sqrt{ \Omega_{HAC}[0, 0] }$$

$$p_{1sided} = 1.0 - F_t(t_\alpha; df = T - 2), \quad p_{2sided} = 2.0 * (1.0 - F_t(|t_\alpha|; df = T - 2))$$

3.3 Andrew Lo (2002) Standalone Sharpe Ratio Inference

To test H0: SR <= 0 against H1: SR > 0 without Gaussianity or IID assumptions, we implement Lo's (2002) influence function formulation under stationary serial correlation:

Influence Function Representation (Proposition 2, Lo 2002):

$$u_t = (r_t - mean(r)) / std(r), \quad SR_{daily} = mean(r) / std(r)$$

$$z_t = u_t - 0.5 * SR_{daily} * (u_t^2 - 1.0)$$

Long-Run Spectral Variance of Influence Function:

$$\Omega_z = \gamma_z(0) + 2.0 * \sum_{j=1}^{L} w(j, L) * \gamma_z(j), \quad \gamma_z(j) = Cov(z_t, z_{t-j})$$

$$SE_{HAC}(SR_{ann}) = \sqrt{ (252.0 / T) * \Omega_z }$$

$$z_{stat} = SR_{ann} / SE_{HAC}(SR_{ann}) \sim N(0, 1)$$

$$p_{1sided} = 1.0 - \Phi(z_{stat}), \quad p_{2sided} = 2.0 * (1.0 - \Phi(|z_{stat}|))$$

3.4 Olivier Ledoit & Michael Wolf (2008) Comparative Sharpe Difference

Comparing two correlated Sharpe ratios evaluated on identical market data requires accounting for contemporaneous covariance and co-kurtosis. Ledoit and Wolf formulate the bivariate Delta method test:

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Parameter Vector & Difference Function:
θ = (μ_1, μ_2, γ_1, γ_2)^T, where γ_i = E[r_i^2], σ_i = sqrt(γ_i - μ_i^2)
ΔSR_daily = f(θ) = (μ_1 / σ_1) - (μ_2 / σ_2), ΔSR_ann = ΔSR_daily * sqrt(252.0)
Analytical Gradient Vector (grad_f in R^4):
grad_f = [ γ_1 / σ_1^3, -γ_2 / σ_2^3, -0.5 * μ_1 / σ_1^3, 0.5 * μ_2 / σ_2^3 ]^T
Spectral Covariance Matrix Ψ in R^{4x4} (u_t = y_t - mean(y)):
Ψ = Γ_0 + sum_{k=1}^L w(k, L) * (Γ_k + Γ_k^T), Γ_k = (1 / T) * sum_{t=k+1}^T u_t * u_{t-k}^T
Var(ΔSR_daily) = (1 / T) * (grad_f^T * Ψ * grad_f), SE(ΔSR_ann) = sqrt( 252.0 * Var(ΔSR_daily) )
z_LW = ΔSR_ann / SE(ΔSR_ann) ~ N(0, 1), p_1sided = 1.0 - Φ(z_LW), p_2sided = 2.0 * (1.0 - Φ(|z_LW|))
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3.5 Politis & Romano (1994) Stationary Block Bootstrap Resampling

Non-parametric resampling is executed using the Stationary Block Bootstrap with wrap-around circular boundaries. Blocks have random geometric length L ~ Geom(p = 1/b), ensuring the resampled time series is strictly stationary.

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Resampling Algorithm (Seed = 42, B = 2,000, b = 21 days):
At step t, continue current block with prob 1 - p (index = (prev + 1) mod N), or start new block with prob p.
Synchronous row resampling preserves contemporaneous cross-correlation across strategy and benchmark vectors.
Studentized Null-Centered Empirical p-values (Davison & Hinkley 1997):
z^{*(b)} = (θ^{*(b)} - θ_hat) / SE_hat, z_base = (θ_hat - 0.0) / SE_hat
p_boot_1sided = ( 1.0 + sum_{b=1}^B I( z^{*(b)} >= z_base ) ) / ( B + 1.0 )
p_boot_2sided = ( 1.0 + sum_{b=1}^B I( |z^{*(b)}| >= |z_base| ) ) / ( B + 1.0 )
Percentile Confidence Intervals (Uncentered):
CI_95% = [ Percentile(θ^*, 2.5), Percentile(θ^*, 97.5) ]
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Methodological Disclaimer: The Stationary Block Bootstrap preserves local serial correlation and volatility clustering up to the block length. It completely severs the Markov chain at block boundaries and does NOT preserve global HMM transition dynamics.

3.6 Multiple-Testing Error Control Architecture

To eliminate family-splitting and guarantee exact global error control, we implement a Unified Primary Omnibus Family alongside an exploratory screening sweep:

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1. Unified Primary Omnibus Family (M = 12 Core Full-Period Hypotheses):
Contains 4x NIFTY 50 Alphas, 4x Equal-Weight Alphas, and 4x Naive Momentum Delta-Sharpes.
Controlled via Holm-Bonferroni step-down at FWER alpha <= 0.05:
p_(k) <= alpha / (M - k + 1), k = 1, ..., M --> Guarantees Global FWER <= 0.05 across all claims.
2. Two-Tier Hierarchical Gatekeeper:
Tier 1 Gatekeeper: NIFTY 50 Alpha (M=4, Holm alpha=0.05). If rejected, proceed to Tier 2 (EW & Momentum, M=8).
3. Exploratory Subperiod Stability Sweep (M = 36 Hypotheses):
Evaluates 4 strategies x 3 comparators x 3 periods. Screened via Benjamini-Hochberg (BH) FDR at q = 0.05:
p_(k) <= (k / M) * q. Benjamini-Yekutieli (BY) sensitivity penalty: c(M) = sum_{m=1}^M (1/m) = 4.168.
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